Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: 4

Start date: 02-02-2026

Names:

S.No	Name
1.	Rohit Rathod
2.	Akshata Chikte
3.	Mounisha Pentakota
4.	Momina Shaik

Internship Duration: 8 weeks

Project Title

INVENTRA – Intelligent Inventory Management System

1. Project Objective

- **Centralized Digital Monitoring:** Design and develop a centralized intelligent pharmacy inventory system to digitally monitor, control, and optimize medical stock handling processes.
- **Real-Time Tracking:** Enable real-time medication tracking to provide accurate visibility of drug availability, batch movement, and usage patterns.
- **Error Reduction:** Minimize human errors, medication mismatches, and data redundancy by replacing manual pharmacy record-keeping with an automated digital solution.
- **Secure Access Control:** Implement role-based authentication and authorization to ensure secure and controlled access for pharmacy administrators, managers, and staff.
- **Traceable Transactions:** Develop a transaction management module to record drug issuance, returns, and stock updates with complete traceability for audit purposes.

2. Project description in detail

The **Smart Hospital Pharmacy Inventory System** is a specialized digital platform designed to modernize and streamline the complex workflows of a medical pharmacy. In many healthcare facilities, medication stock is still managed using manual registers or basic spreadsheets, which often leads to critical issues such as stock mismatches, delayed updates, data redundancy, and a dangerous lack of real-time visibility into life-saving drug availability. This system addresses these high-stakes challenges by providing a centralized, automated, and secure inventory management solution tailored for medical environments.

The system allows authorized pharmacy personnel to manage medical items with complete control over stock records. Each medication entry includes detailed attributes such as the drug name, category (e.g., Antibiotics, Analgesics), quantity, supplier information, and current stock status. Through an intuitive interface, users can easily add new medicines, update stock levels, delete outdated records, and perform rapid searches for specific products, ensuring efficient day-to-day pharmacy operations.

To maintain strict data security and controlled access, the system incorporates a role-based authentication mechanism. Administrators (such as Chief Pharmacists) can manage users and oversee the entire pharmacy inventory, while staff members (such as Pharmacy Technicians) are restricted to permitted operations like stock updates and recording drug issuance. This hierarchical structure ensures both professional accountability and the protection of sensitive medical data within the hospital system.

A core feature of the platform is its transaction tracking module, which meticulously records all stock movements, including medicine issuance to departments and patient returns. Every transaction is logged with precise timestamps and user details, enabling full traceability for medical audits and reducing the risk of stock discrepancies or medicine misuse. The system automatically updates inventory quantities after each transaction is finalized, ensuring that stock information remains accurate and up to date in real time.

An interactive dashboard provides real-time visibility into stock counts, usage trends, and category distribution through visual charts. This data-driven approach enables pharmacy managers to monitor frequently prescribed medicines and optimize restocking cycles efficiently.

3. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis, pharmacy workflow identification, and system architecture design.	Conducted detailed study of hospital pharmacy challenges, finalized medicine tracking requirements, and designed module breakdown.
Week 2	Database schema design and backend pharmacy environment setup.	Designed normalized relational database schema for medicines, users, and transactions; configured Spring Boot environment.
Week 3	Authentication system and role-based access control.	Implemented secure login system, user registration, and role-based authorization for Pharmacy Admins and Staff.
Week 4	Pharmacy management module development.	Developed core features including medicine addition, stock updates, category classification, and rapid search functionality.
Week 5	Drug transaction management system.	Implemented medicine issue and return tracking with automatic stock adjustments and complete audit history logging.
Week 6	Dashboard development and data visualization.	Built interactive dashboard displaying total stock, category-wise distribution, low-stock alerts, and usage statistics.
Week 7	System testing, debugging, and performance optimization.	Performed unit and integration testing, resolved functional errors, and optimized database queries for medical records.
Week 8	Final system integration, documentation, and deployment.	Integrated all pharmacy modules, prepared technical guides, conducted final demo, and completed project submission.

4. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial project briefing, requirement analysis, pharmacy system planning, and technology stack finalization	07-02-26
Database & Backend Setup Completed	Successful design of database schema for medical records and implementation of core backend structure with API connectivity	Week-1
Authentication Module Completed	Implementation of secure login system with role-based access control for Pharmacy Admin and Staff users	Week-2
Pharmacy Module Prototype Ready	Development of core inventory features including medicine management, category classification, and stock updates	Week-3

Transaction Tracking System Completed	Implementation of drug issue/return logging with automatic stock adjustment and history tracking for audits	Week-4
Dashboard & Analytics Integration	Deployment of interactive dashboard with medical stock insights, graphical charts, and alert indicators	Week-5
System Testing & Optimization Completed	Completion of functional testing, medical data validation, and performance improvements	Week-6
Final System Deployment & Documentation	Full system integration, preparation of technical pharmacy documentation, and successful project demonstration	Week-7

5. Project execution details

The **Smart Hospital Pharmacy Inventory System** is engineered as a high-availability full-stack application, prioritizing data consistency and rapid response times essential for clinical environments.

Frontend Architecture (Next.js & TypeScript)

- **Client-Side Dynamics:** The user interface leverages **React.js (Next.js)** to implement a Single Page Application (SPA) architecture, ensuring smooth navigation without page reloads during critical pharmacy operations.
- **Type-Safe Development:** By utilizing **TypeScript**, the team enforced strict data structures for medication entities, batch numbers, and stock counts, significantly reducing runtime errors and ensuring data integrity before it reaches the backend.
- **Responsive Styling: Tailwind CSS** was used to create a utility-first, mobile-responsive design, allowing the pharmacy portal to be accessed seamlessly on both desktop workstations and handheld tablets during stock-taking.

Backend & API Logic (Node.js & Next.js Routes)

- **Serverless API Routes:** The backend logic is encapsulated within **Next.js API routes (Node.js)**, which handle RESTful requests for drug issuance, returns, and user management.
- **Business Logic Layer:** This layer processes complex pharmaceutical workflows, such as calculating quantity deductions and validating that medicine is not issued beyond available stock levels.
- **Optimized Response Handling:** The API structure is designed to minimize latency, ensuring that pharmacists receive instant confirmation for transactions and stock searches.

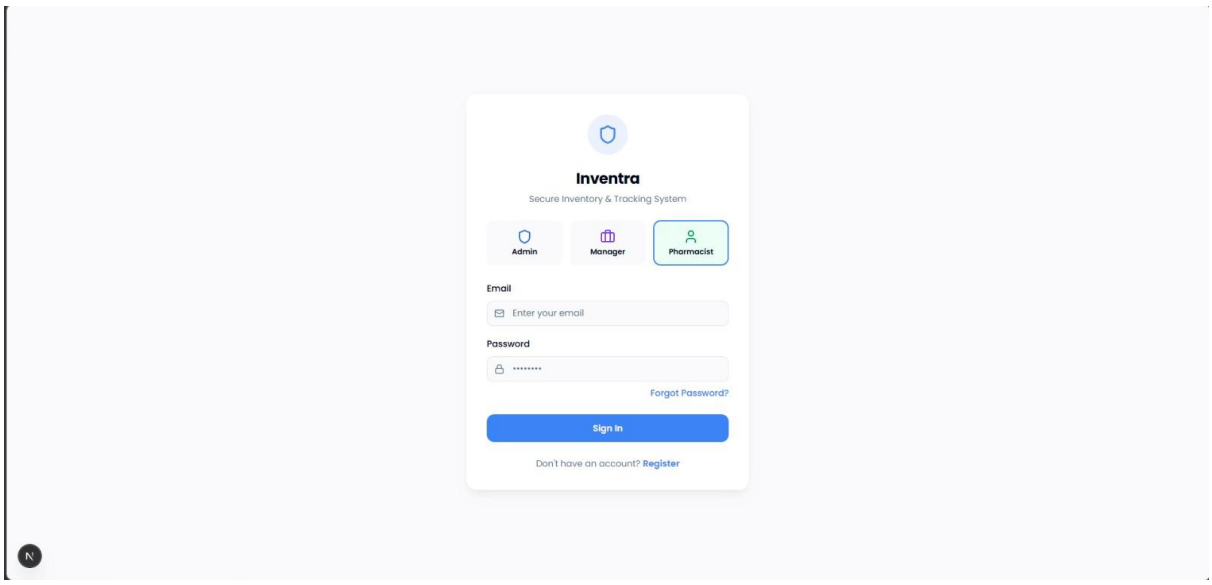
Cloud Data Management (Firebase)

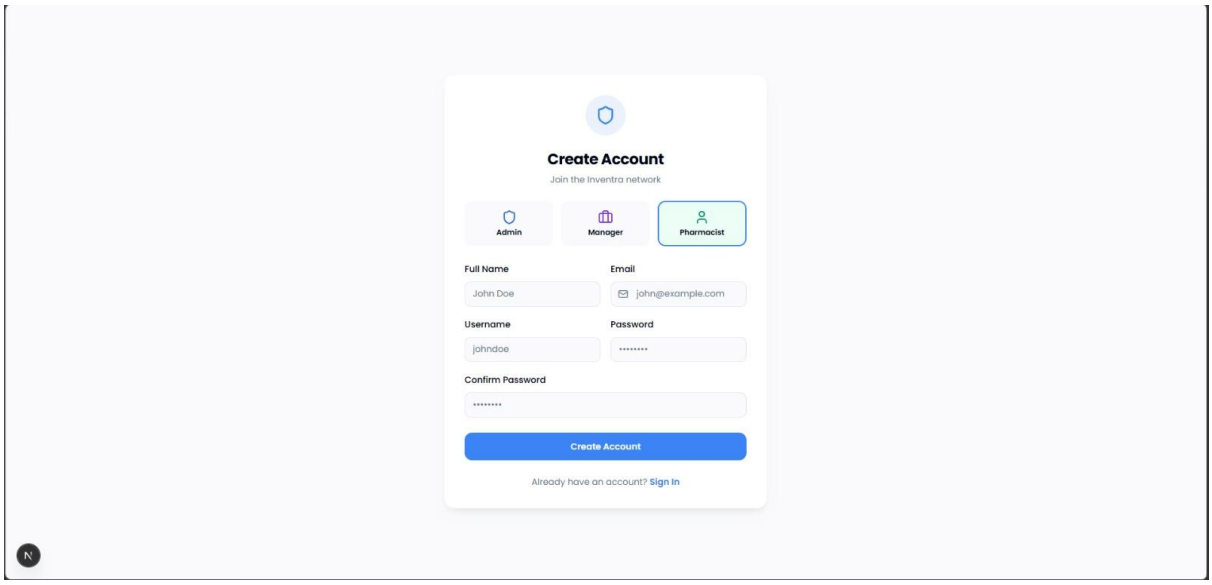
- **Real-time Synchronization: Firebase (Cloud Firestore / Realtime Database)** serves as the backbone for data persistence. Its WebSocket-based listeners ensure that stock updates at one terminal are broadcast to all other active sessions in milliseconds.
- **NoSQL Flexibility:** The database schema uses a document-oriented approach to store varied medicine attributes—including dosage, manufacturer, and category—allowing the system to scale as new types of medical supplies are added.
- **Cloud Security:** Firebase Security Rules were implemented to provide an additional layer of protection at the database level, ensuring data can only be modified by authenticated users with appropriate roles.

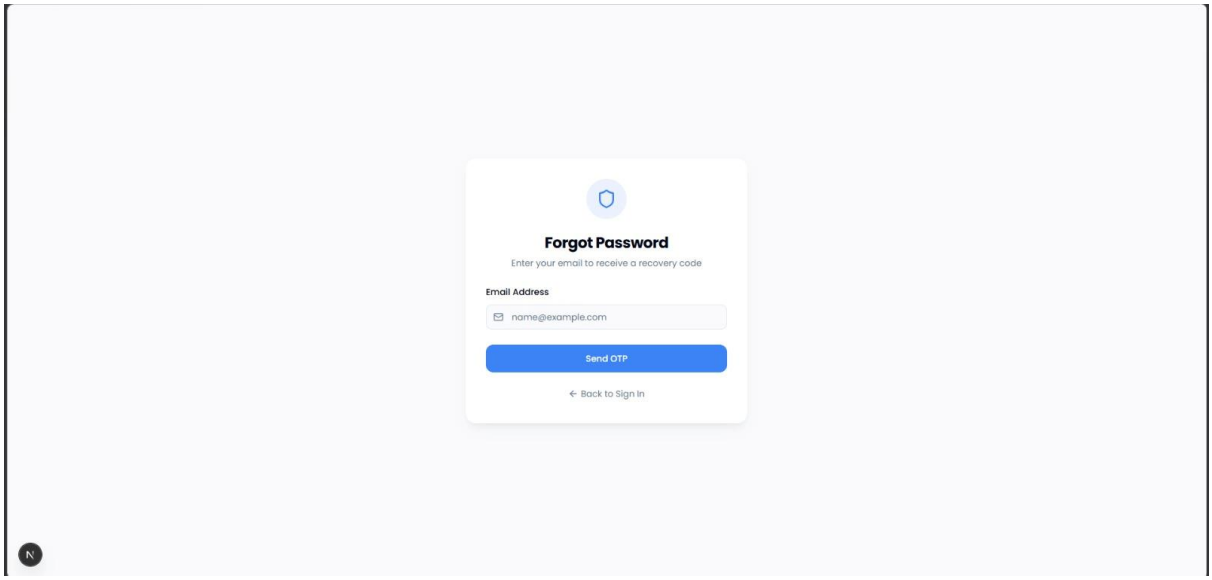
Security & Auditability

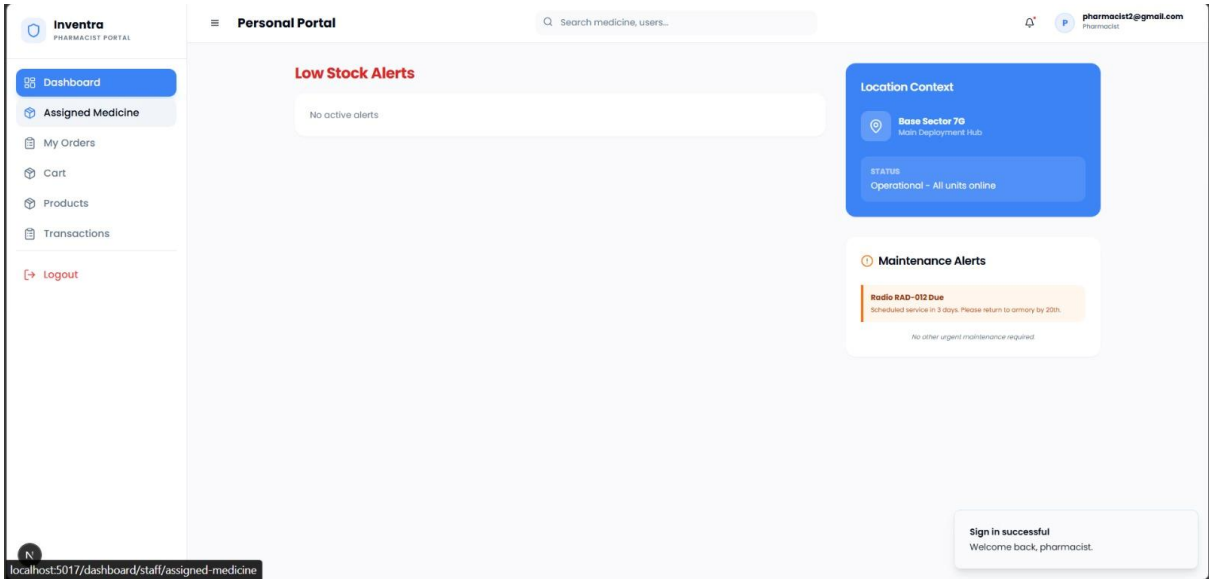
- **Identity Management:** The system integrates **Firebase Authentication** for secure sign-in, utilizing industry-standard encryption for user credentials.
- **Role-Based Access Control (RBAC):** Permissions are strictly bifurcated; while **Pharmacy Administrators** have full CRUD (Create, Read, Update, Delete) access to inventory and user accounts, **Staff** are restricted to transactional operations like drug issuance and stock logging.
- **Comprehensive Audit Trail:** Every medication movement triggers a background log event in the transaction tracking module, recording the **Unique Transaction ID**, **User ID**, **Medicine ID**, and an **Immutable Server Timestamp** to satisfy medical legal requirements

6. Snapshots / Screenshots






Forgot Password
Enter your email to receive a recovery code
Email Address
name@example.com
Send OTP
← Back to Sign In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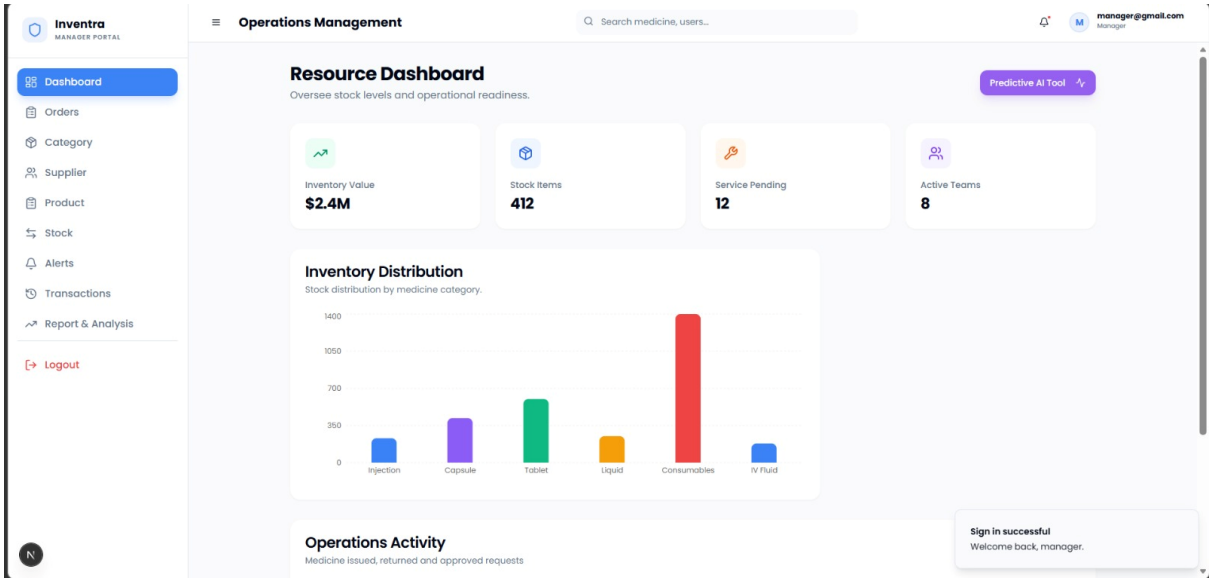
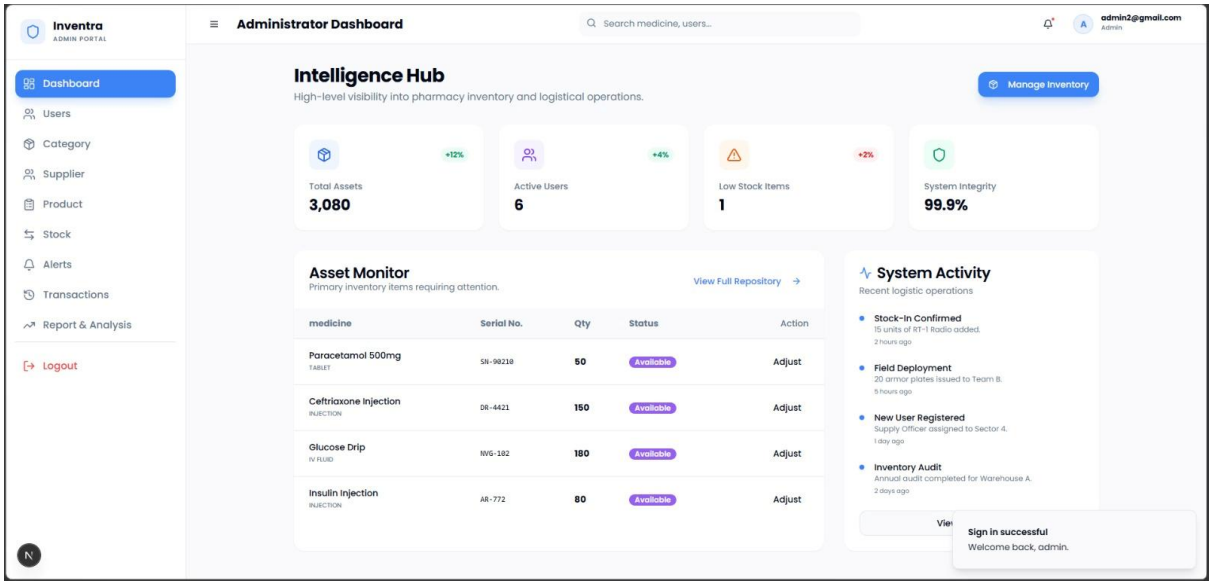
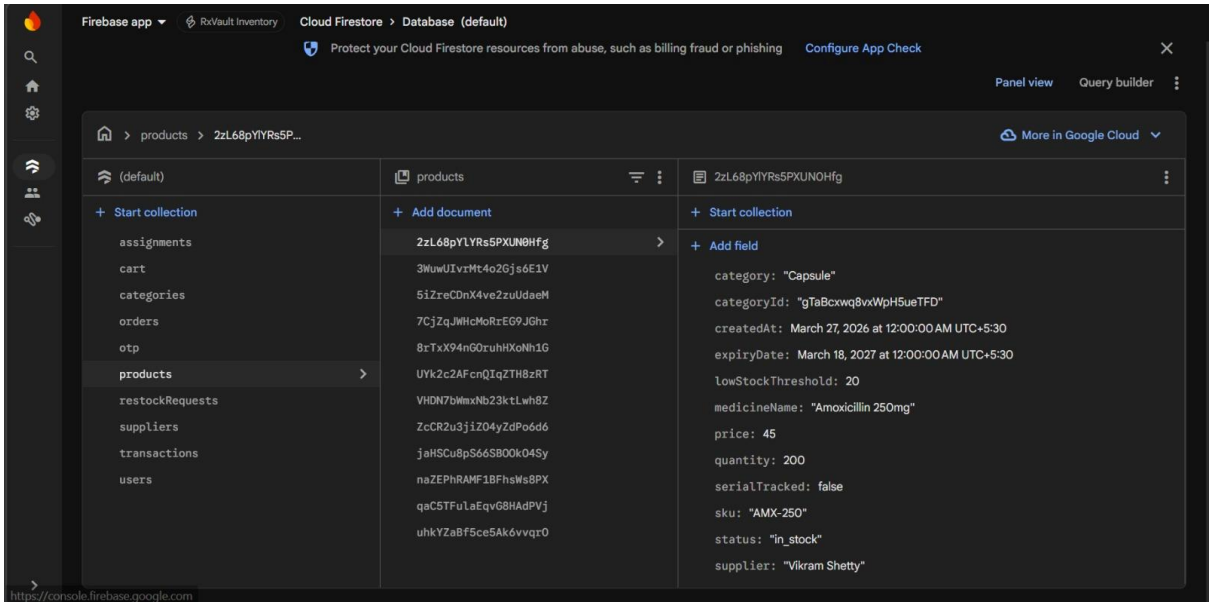
PHARMACIST PORTAL
Personal Portal
Search medicine, users...
pharmacist2@gmail.com
Pharmacist
Dashboard
Assigned Medicine
My Orders
Cart
Products
Transactions
Logout
Low Stock Alerts
No active alerts
Location Context
Base Sector 7G
Main Deployment Hub
STATUS
Operational - All units online
Maintenance Alerts
Radio RAD-012 Due
Scheduled service in 3 days. Please return to armory by 20th.
No other urgent maintenance required
Sign in successful
Welcome back, pharmacist.
localhost:5017/dashboard/staff/assigned-medicine


Inventra
MANAGER PORTAL
Operations Management
Search medicine, users...
manager@gmail.com
Manager
Dashboard
Orders
Category
Supplier
Product
Stock
Alerts
Transactions
Report & Analysis
Logout
Resource Dashboard
Oversee stock levels and operational readiness.
Predictive AI Tool
Inventory Value
$2.4M
Stock Items
412
Service Pending
12
Active Teams
8
Inventory Distribution
Stock distribution by medicine category.
1400
1050
700
350
0
Injection
Capsule
Tablet
Liquid
Consumables
IV Fluid
Operations Activity
Medicine issued, returned and approved requests
Sign in successful
Welcome back, manager.





7. Learnings & Skills Acquired

- **Relational Database Management:** Gained practical experience in designing and managing complex relational databases using MySQL specifically for medical records.
- **Backend Development:** Developed a strong understanding of backend development using Java and Spring Boot for building secure, high-performance RESTful APIs.
- **Security Implementation:** Learned to implement secure authentication and role-based authorization mechanisms to protect sensitive healthcare data.
- **Frontend Engineering:** Acquired skills in frontend development using React.js to build responsive, user-friendly interfaces suitable for fast-paced hospital environments.
- **System Integration:** Improved knowledge of API integration and seamless data exchange between frontend and backend systems.
- **Domain Logic:** Understood the real-world implementation of inventory management logic, focusing on critical medical stock control and safety protocols.

- **Data Visualization:** Learned to design and display interactive dashboards and data visualizations to assist in pharmaceutical decision-making.
- **Quality Assurance:** Enhanced ability in debugging, unit testing, and optimizing system performance to ensure medical data reliability.
- **Full-Stack Architecture:** Gained hands-on experience in end-to-end application architecture and the integration of multiple software modules.
- **Professional Skills:** Improved team collaboration, technical documentation, and project presentation skills within a structured development lifecycle.

8. Challenges Faced

Challenge 1: Designing an Efficient Database for Medical Records

- **Problem:** Managing complex relationships between pharmacy users, thousands of medicine SKUs, and frequent transaction logs without data redundancy.
- **Solution:** Designed a normalized MySQL schema with proper foreign keys and indexing to maintain data integrity and improve query performance for rapid medicine lookups.

Challenge 2: Maintaining Accurate Real-Time Medical Stock Updates

- **Problem:** Ensuring medication quantities remain 100% correct after every issue and return transaction to prevent dangerous stockouts of life-saving drugs.
- **Solution:** Implemented robust backend logic in Java (Spring Boot) to automatically update stock levels and validate quantity constraints during every transaction.

Challenge 3: Implementing Secure Role-Based Access Control (RBAC)

- **Problem:** Preventing unauthorized personnel from accessing or modifying sensitive pharmacy inventory data and controlled substance records.
- **Solution:** Developed a secure role-based authentication system using encrypted passwords and controlled API access to differentiate between Admin and Staff permissions.

Challenge 4: Integrating React Frontend with Java Backend for Medical Speed

- **Problem:** Handling smooth, high-speed data exchange between the frontend and backend services to ensure pharmacists can act quickly during emergencies.
- **Solution:** Built optimized RESTful APIs and implemented comprehensive error handling and response management for seamless system integration.

9. Testimonials from team

Working on the **Smart Hospital Pharmacy Inventory System** was a valuable learning experience for the entire team. Each member contributed to different modules such as backend development, frontend design, and database management, which helped us understand how a real-world full-stack system is built and integrated for a critical healthcare environment.

Regular discussions, planning sessions, and collaborative debugging helped us overcome technical challenges efficiently. The project significantly improved our practical knowledge in **Java**, **MySQL**, and **React**, while also strengthening our problem-solving abilities in handling sensitive medical data. This experience enhanced our teamwork, communication, and project documentation skills, making us more confident in handling industry-level software development projects.

10. Conclusion

The **Smart Hospital Pharmacy Inventory System** successfully demonstrates how digital technology can transform traditional medical inventory handling into a smart, efficient, and data-driven process. By integrating a **Java-based backend**, a **MySQL relational database**, and a **React frontend**, the system delivers a complete full-stack solution for real-time pharmacy monitoring and control.

The project effectively addressed common challenges found in manual pharmacy systems, such as medication stock mismatches, lack of transparency in drug movement, and delayed record updates. Through critical features like **role-based authentication**, **transaction tracking**, **automated stock updates**, and **visual dashboards**, the system ensures accuracy, accountability, and ease of management.

Developing this system provided hands-on experience in backend API development, database design, frontend integration, and system testing. It also strengthened our team's understanding of how different components of a software system interact to create a reliable and scalable healthcare application. Overall, this project stands as a practical and scalable solution that can be adapted for hospitals and clinics requiring efficient medical inventory control. The project not only met its technical objectives but also enhanced our readiness to work on real-world software development challenges.

11. Acknowledgements

We would like to express our sincere gratitude to our project mentor and faculty members for their continuous guidance, encouragement, and valuable feedback throughout the development of the Inventra – Intelligent Inventory Management System. Their technical insights and support played a crucial role in helping us overcome challenges and improve the quality of our work.

The opportunity to work on a real-time, practical system has significantly enhanced our technical knowledge and confidence. Finally, we appreciate the teamwork, dedication, and collaborative efforts of all team members, whose consistent contributions made the successful completion of this project possible.